

K Level	Bloom's Taxonomy Action Verbs
K1	Choose, Define, Find, How, Label, List, Recall, Relate, Select, Show, Spell, What, When, Where, Which, Who, Why
K2	Classify, Compare, Contrast, Demonstrate, Explain, Extend, Illustrate, Infer, Interpret, Outline, Relate, Rephrase, Show, Summarize, Translate
K3	Apply, Build, Choose, Construct, Develop, Experiment with, Identify, Interview, Make use of, Model, Organize, Plan, Select, Solve, Utilize
K4	Analyze, Assume, Categorize, Classify, Compare, Conclusion, Contrast, Discover, Dissect, Distinguish, Divide, Examine, Function, Inference, Inspect
K5	Appraise, Assess, Award, Choose, Compare, Conclude, Criticize, Decide, Deduct, Defend, Determine, Disprove, Estimate, Evaluate, Explain, Importance, Influence, Interpret, Judge, Justify, Measure, Opinion, Perceive, Prioritize, Prove, Recommend
K6	Adapt, Build, Change, Choose, Combine, Compile, Compose, Construct, Create, Delete, Design, Develop, Discuss, Elaborate, Estimate, Formulate, Happen, Imagine, Improve, Invent, Makeup, Maximize, Minimize, Modify, Original, Originate, Plan, Predict, Propose, Solution, Solve, Suppose, Test, Theory

Unit IV

Processor

Course Objective – COB4: To Understand the techniques involved in Processor

Course Outcome - CO4: learn the concepts in Processor.

Bloom's Taxonomy Level: K1,K2,K3

Part A

S.No.	Question	BTL*
1.	What are the key steps involved in instruction execution? The key steps in instruction execution are: • Fetch: Retrieving the instruction from memory. • Decode: Interpreting the instruction. • Execute: Performing the operation. • Memory Access: Accessing data if required. • Write Back: Storing the result back in the register or memory.	K1
2.	What is a data path in a CPU? A data path is the part of the CPU that performs operations on data. It includes components such as: • Registers: Temporary storage. • ALU (Arithmetic Logic Unit): For arithmetic and logical operations. • Multiplexers: Directing data flow. • Control signals: Directing how data flows through the system.	K1
3.	What is the role of a control unit in a CPU? The control unit generates control signals that manage the execution of instructions. It decodes instructions, and orchestrates the operations of the CPU's data path, memory, and input/output devices to ensure correct execution.	K1
4.	Define pipelining and its significance. Pipelining is a technique where multiple instruction phases (fetch, decode, execute, etc.) are overlapped to improve performance. It allows the CPU to work on several instructions simultaneously, increasing throughput and reducing execution time.	K1
5.	What are RAW, WAR, and WAW hazards? These are types of data hazards: • RAW (Read After Write): Occurs when a read depends on a write not yet completed. • WAR (Write After Read): Occurs when a write happens before a read is completed. • WAW (Write After Write): Occurs when two writes are done out of order.	K1
6.	What is branch prediction? Branch prediction is a technique used in pipelined CPUs to guess the outcome	K1

	of a branch instruction (whether a condition is true or false) and continue execution along that path. Correct prediction avoids pipeline stalls and improves efficiency.	
7.	What is a control hazard in pipelining? Control hazards occur when the pipeline is uncertain about the next instruction to fetch due to branch instructions (conditional jumps). This can cause delays, as the correct path of execution is not immediately known.	K1
8.	What are the main components of a pipelined architecture? Answer: The main components of a pipelined architecture include: • Instruction Fetch (IF): Fetches the next instruction. • Instruction Decode (ID): Decodes the instruction and reads operands. • Execute (EX): Executes the instruction. • Memory Access (MEM): Accesses memory for load/store. • Write Back (WB): Writes results back to registers.	K1
9.	How does forwarding resolve data hazards? Forwarding (also called bypassing) resolves data hazards by allowing intermediate results to be used by subsequent instructions without waiting for them to be written back to the register. It reduces pipeline stalls caused by dependencies.	K1
10.	What is speculative execution in the context of control hazards? Speculative execution is a technique where the CPU executes instructions ahead of time based on predicted outcomes (usually for branches). If the prediction is correct, performance is improved; if not, the speculative results are discarded, and execution restarts from the correct point.	K1
12.	What are the two types of control units in a CPU? • Hardwired Control Unit: Uses fixed logic circuits to generate control signals. • Microprogrammed Control Unit: Uses microinstructions stored in control memory to generate control signals.	K1
13.	What is instruction pipelining and how does it differ from non-pipelined execution? Instruction pipelining allows multiple instructions to be processed simultaneously by dividing the instruction execution process into stages. In contrast, non-pipelined execution processes only one instruction at a time, completing one instruction before starting the next.	K1
14.	What is the difference between a control hazard and a data hazard?	K1

	•Data hazard: Occurs when instructions in the pipeline depend on data that hasn't been processed yet. •Control hazard: Arises from the pipeline's uncertainty about the control flow, usually caused by branch or jump instructions.	
15.	How does a pipeline improve CPU performance? A pipeline improves CPU performance by overlapping the execution of multiple instructions. This increases the instruction throughput, allowing the CPU to process more instructions in less time.	K1
16.	What is a register file in the context of a CPU's data path? A register file is a set of registers that store temporary data during instruction execution. It allows quick access to operands required by the ALU and stores results from operations.	K1
17.	How does instruction-level parallelism (ILP) relate to pipelining? Instruction-level parallelism (ILP) refers to the degree to which instructions can be executed in parallel. Pipelining exploits ILP by overlapping different stages of instruction execution, increasing the number of instructions processed simultaneously.	K4
18.	What is instruction decode in the instruction execution cycle? Instruction decode is the stage in the instruction execution cycle where the CPU interprets the fetched instruction, identifies the operation to be performed, and determines the operands involved.	K1
19.	How does a CPU handle a RAW hazard in a pipeline? A CPU handles a Read After Write (RAW) hazard by either stalling the pipeline until the necessary data is available or using techniques like forwarding to pass the required data directly from one instruction to the next.	K4
20.	Define the term 'pipeline depth.' Pipeline depth refers to the number of stages in a pipeline. The more stages, the deeper the pipeline, allowing for finer instruction execution steps but potentially introducing more hazards.	K1

Part B

S.No.	Question	BTL *
1.	Explain in detail about data path design in a CPU? Discuss the components of a typical data path, such as registers, ALU, multiplexers, and memory units.	K3

2.	What are data hazards in a pipelined CPU? Explain the types of data hazards and its techniques.	K3
3.	Discuss the design of a simple 5-stage pipelined processor. Provide a detailed explanation of each stage (IF, ID, EX, MEM, WB) and the interactions between stages. What are the challenges in implementing pipelining, and how are they resolved?	K2
4.	Explain in detail about the control unit in a CPU. Compare and contrast hardwired and microprogrammed control units in terms of their design, performance, and flexibility.	K2
5.	Explain in detail about Instruction Execution in processor?	K2